

Name	CSU ID #
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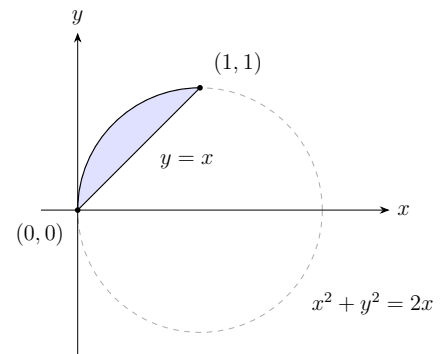
Choose and answer **exactly two** of the four problems. If you attempt more than two, only the first two will be graded. For Questions 1, 2, and 4, only answers in the boxes will be graded. For Question 3, your final labeled sketch in the space below the question will be graded. You may use the surrounding space for scratch work. Each problem is worth the same number of points.

1. Rewrite as a single polar integral in the order $dr d\theta$.

Do not evaluate.

$$\int_0^1 \int_x^{\sqrt{2x-x^2}} (x^2 + y^2) dy dx.$$

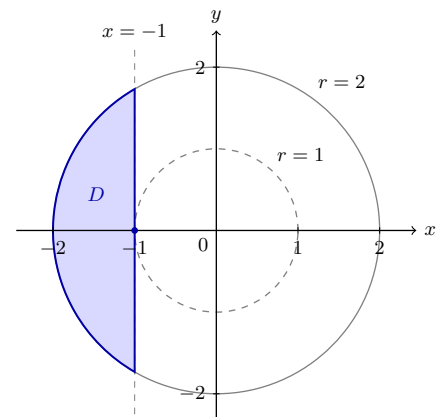
Hint: First write the region's Cartesian bounds explicitly. Then convert those bounds to polar coordinates.



2. The shaded lamina is the part of $1 \leq x^2 + y^2 \leq 4$ with $-2 \leq x \leq -1$. Its density is $\delta(x, y) = x^2 + y^2 - 1$. Use polar coordinates to find its mass.

Hint: First write the region's Cartesian bounds explicitly. Then convert those bounds to polar coordinates. You may use

$$\int \sec^4 \theta d\theta = \tan \theta + \frac{1}{3} \tan^3 \theta + C.$$



3. A solid has cylindrical bounds

$$\begin{cases} 0 \leq \theta \leq 2\pi, \\ 0 \leq r \leq 2, \\ r^2 \leq z \leq 2r. \end{cases}$$

Sketch and shade its cross section in the entire xz -plane ($y = 0$). Label the points where the two boundary curves meet.

4. Write spherical-coordinate bounds for

$$E = \{(x, y, z) : 1 \leq x^2 + y^2 + z^2 \leq 4, x \leq 0, y \geq 0\}.$$

Here φ is measured from the positive z -axis, and θ is the azimuthal angle measured from the positive x -axis in the xy -plane.

